GigaWhat Energy

GigaWhat Electricity
Controlled procedure

PR-ELEC-012 Isolation and Earthing of 11 kV Circuits

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Owner	Head of Electricity Network Operations
Approved by	Electricity Safety Manager
Applies to	All GigaWhat Electricity sites: 11 kV switchgear, ring main units and feeders

FICTIONAL DOCUMENT. Created for the GigaWhat demo. Not for real operational use.

1 Purpose

This procedure sets out how 11 kV circuits are isolated, proved dead and earthed before a Permit-to-Work is issued under PR-ELEC-020, so that nobody works on 11 kV equipment that is live or could become live.

2 Scope

This procedure applies to 11 kV switchgear, ring main units and feeders at all GigaWhat Electricity sites, including SWG-ASH-11 (ESS-ASH), SWG-KGM-11 (ESS-KGM), RMU-BRK-01 (ESS-BRK), RMU-ORC-01 (ESS-ORC) and feeders FDR-ASH-07, FDR-ASH-09 and FDR-KGM-03. It does not cover 33 kV, 132 kV or low voltage equipment.

3 References

- PR-ELEC-001 Distribution Safety Rules - Summary for Operational Staff
- PR-ELEC-015 Switching Programmes on the 11 kV and 33 kV Networks
- PR-ELEC-020 Permit-to-Work and Sanction-for-Test (Electrical)
- PR-ELEC-075 Fault Restoration on 11 kV Feeders
- PR-CORP-001 Reporting Incidents and Near Misses
- HSE HSG85 Electricity at work: Safe working practices

- HSE HSG253 The safe isolation of plant and equipment
- HSE HSR25 Memorandum of guidance on the Electricity at Work Regulations 1989
- Electricity at Work Regulations 1989

4 Definitions

Point of isolation: a place where equipment is disconnected from a source of supply by an open switching device secured in the open position.

Circuit Main Earth (CME): an earth applied at a point of isolation by an Authorised Person (HV) before a permit is issued, and left in place until the permit is cancelled.

Additional Earth: an earth applied at the point of work by the Competent Person (HV) after the permit is issued.

Approved voltage indicator: a voltage indicator of a type approved by the Electricity Safety Manager for 11 kV, with a current calibration label.

Safety lock: a lock with a unique key, used only to secure points of isolation and earthing devices.

Safety Key Safe: a lockable box at site in which safety lock keys are kept while the permit is in force.

Caution notice: a notice warning that equipment must not be operated or interfered with.

Danger notice: a red notice stating that equipment is isolated for work and must not be operated.

5 Responsibilities

Senior Authorised Person (HV): approves isolations involving more than two sites or 33 kV plant, and resolves any doubt about points of isolation.

Authorised Person (HV): plans the isolation, carries out the switching, proves dead, applies Circuit Main Earths, secures each point of isolation and issues the permit.

Competent Person (HV): receives the permit, applies and records any Additional Earths, and carries out the work.

Control Engineer: approves the switching programme, authorises each operation and keeps the network diagram up to date with every point of isolation, lock and earth.

6 Procedure

6.1 Planning the isolation

6.1.1 Identify the equipment to be worked on by its asset ID and circuit name, and confirm the scope of work.

6.1.2 Identify every point from which the equipment could be made live, including both ends of a feeder, normally open points, ring main unit switches, customer generation and backfeed from the low voltage network through distribution transformers.

> NOTE: FDR-ASH-07 runs between SWG-ASH-11 at ESS-ASH and SWG-KGM-11 at ESS-KGM. Both ends are points of isolation, and both must be isolated and earthed.

6.1.3 Prepare a switching programme in accordance with PR-ELEC-015 that lists every point of isolation and every Circuit Main Earth.

6.1.4 Submit the switching programme to the Control Engineer for checking against the network diagram and approval.

6.1.5 Obtain the approval of the Senior Authorised Person (HV) where the isolation involves more than two sites or 33 kV plant.

6.1.6 Before travelling to site, check that you have an approved voltage indicator, a proving unit, the correct earthing equipment, safety locks and notices.

6.2 Isolating the circuit

6.2.1 Before each switching operation, contact the Control Engineer, state the operation to be carried out and repeat back the instruction given.

> WARNING: The oil circuit breakers on SWG-KGM-11 are under an operational restriction. A failure while operating them can cause an oil fire or explosion in the switchroom, so they must be operated remotely as set out in PR-ELEC-015.

6.2.2 Open the circuit breaker or switch at each point of isolation.

6.2.3 Withdraw the circuit breaker to the isolated position, or open the isolating switch, so that there is a visible or positively indicated break.

6.2.4 Where spout shutters are fitted, close the busbar and circuit shutters and lock them closed.

6.2.5 Where a locking facility is fitted, lock the point of isolation with a safety lock and place the key in the Safety Key Safe.

6.2.6 Fix a Caution notice or Danger notice at each point of isolation.

6.2.7 Report each completed operation to the Control Engineer, who records it on the network diagram.

6.3 Proving dead

> WARNING: A voltage indicator can fail without warning. If a faulty indicator is used, live equipment will appear dead and anyone who touches it may be killed. The indicator must be proved immediately before every test for dead.

6.3.1 Check that the approved voltage indicator is suitable for 11 kV, is within its calibration date and is undamaged.

6.3.2 Prove the voltage indicator on a known live source or proving unit immediately before testing for dead.

6.3.3 Test each phase for dead at the point where the Circuit Main Earth will be applied.

6.3.4 If the voltage indicator fails to indicate when it is proved, do not use it, and repeat 6.3.2 and 6.3.3 with a different approved voltage indicator.

6.3.5 Remove a failed voltage indicator from use, label it as defective and report the failure under PR-CORP-001.

6.3.6 Where a ring main unit such as RMU-BRK-01 or RMU-ORC-01 has no test access, check the integral voltage presence indicators on all three phases and prove the circuit dead by closing the fault-making earth switch, with the agreement of the Control Engineer.

6.4 Applying Circuit Main Earths

> WARNING: Applying an earth to a conductor that is still live causes a flashover that can kill or seriously burn the operator. Earths must only be applied after the circuit has been proved dead under 6.3.

6.4.1 Apply Circuit Main Earths at the points of isolation where practicable.

6.4.2 Use the integral earth switch where one is fitted. Where none is fitted, use approved portable earthing equipment, connecting the earth end first and the phase ends last.

6.4.3 Lock each earth switch used as a Circuit Main Earth in the earthed position with a safety lock and place the key in the Safety Key Safe.

6.4.4 Record the location of the Circuit Main Earths on the permit.

6.4.5 Report each Circuit Main Earth to the Control Engineer, who shows it on the network diagram.

6.5 Issuing the permit

> HOLD POINT: The Authorised Person (HV) must not issue the permit until they have confirmed that the circuit has been isolated, proved dead and earthed, and that the safety precautions are recorded on the permit.

6.5.1 Obtain a permit number from the Control Engineer and confirm that the network diagram matches the isolation and earthing on site.

6.5.2 Issue the Permit-to-Work in accordance with PR-ELEC-020.

6.5.3 Show the Competent Person (HV) each point of isolation, each Circuit Main Earth and the point of work, and identify any adjacent equipment that remains live.

6.5.4 Lock the Safety Key Safe and keep its key with the Authorised Person's copy of the permit until the permit is cancelled.

6.6 Work at the point of work

6.6.1 As the Competent Person (HV), prove each conductor dead at the point of work in accordance with 6.3 before touching it.

6.6.2 Apply Additional Earths at the point of work where the work requires them, and record them on the permit.

> WARNING: An 11 kV cable that has been wrongly identified may still be live. Cutting into it causes an explosive fault that can kill the person cutting it.

6.6.3 Before cutting an 11 kV cable, identify it with approved cable identification equipment and spike it with the approved remote cable spiking tool.

6.6.4 Do not remove or move a Circuit Main Earth. Removal for testing requires a Sanction-for-Test under PR-ELEC-020.

6.7 Removing earths and restoring the circuit

6.7.1 As the Competent Person (HV), remove all Additional Earths, tools and materials, withdraw the working party and clear the permit.

6.7.2 As the Authorised Person (HV), check the point of work and cancel the permit in accordance with PR-ELEC-020.

> HOLD POINT: No Circuit Main Earth may be removed until the permit has been cancelled and the Control Engineer has confirmed the next step of the switching programme.

6.7.3 Remove the Circuit Main Earths in the order given in the switching programme, reporting each removal to the Control Engineer.

6.7.4 Remove the safety locks and notices, return the keys from the Safety Key Safe, and restore the circuit as instructed by the Control Engineer.

6.7.5 Return the completed switching programme to the Control Engineer.

6.8 Emergency

6.8.1 If anyone is injured or there is a fire, call 999 and then inform the Control Engineer.

6.8.2 If someone has received an electric shock, do not touch them until the supply has been isolated or they have been moved clear by insulated means.

6.8.3 If a member of the public reports a power cut or damaged equipment, ask them to call 105.

6.8.4 Make the site safe, preserve the scene and report the event under PR-CORP-001.

7 Records

- Signed switching programme, kept by the Control Engineer for six years.
- Permit-to-Work, kept under PR-ELEC-020.
- Control log and network diagram entries.
- Voltage indicator calibration records, kept by the Electricity Safety Manager.

8 Revision history

Version	Date	Change
1	2016-05-03	First issue, replacing the local isolation instructions for Ashford Road and Kingsmead.
2	2019-06-03	Added proving dead at ring main units without test access, and remote cable spiking before cutting.
3	2023-02-01	Restructured to align with PR-ELEC-020 v6 and PR-ELEC-015 v4. Added repeat-back of switching instructions to the Control Engineer.